

The effect of arm swing on lower extremities in vertical jumping

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Abstract

Although it is known that an arm swing can enhance the performance in vertical jumping, the mechanisms through which this enhancement occurs are not yet clearly described. The purpose of this study was to examine how arm swing affects the lower extremity torque, power and work in vertical jumping and to gain an insight into the mechanisms that enable the arm swing to increase jump height. Five subjects maximally performed two types of vertical squat jumps with (SJA) and without (SJ) an arm swing from a force platform. All performances were videotaped with a high-speed video camera (200 Hz). The jump heights, joint torques, power and work were calculated by combining kinematic and kinetic data. It was confirmed that arm swing enhanced the jump height significantly ($p < 0.01$). The work by the hip and by the ankle was significantly augmented by arm swing ($p < 0.05$ and $p < 0.01$, respectively). However, the work by the knee was significantly smaller in SJA ($p < 0.05$). The total work by the three lower extremity joints (ankle, knee and hip) was significantly larger in SJA ($p < 0.05$). The increase of the lower extremity work by the arm swing (31.4 J) was about twice as large as the work done by the shoulder and elbow in SJA (16.3 J). It was concluded that the increment of jump height resulted mainly from the increase of the lower extremity work, which is considered to have been brought about by the additional load on the lower extremity due to the arm swing.

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1. Introduction

Since man is bipedal, free arm movements are important for maintaining balance and moving effectively. From this point of view, the effects of arm swings on jump performance have been investigated within the past few decades. Several researchers (Payne et al., 1968; Miller, 1976; Shetty and Etnyre, 1989; Harman et al., 1990) investigated arm swing jumps using force platforms. These studies showed that the ground reaction force (F_{GR}) in the latter half of the propulsive phase was increased by the use of the arms. This increase in the F_{GR} enhanced the net ground reaction impulse. This raised the position and vertical velocity of the center of

mass (CM) at takeoff, which increased the jump height. The change in F_{GR} due to arm swing implies that the arm swing has some effect on the movements of the body segments central to producing a jumping motion, i.e., the lower extremities. Luhtanen and Komi (1978) studied the separate segmental movements at maximum intensity to quantify the contribution of the different body segments including the arms. They suggested that the arm swing contribution to the takeoff velocity was 10%. However, their definition of “segmental performances” failed to accurately define each joint movement. Therefore the calculated contribution should be considered with caution. In addition, mere summation of independent segmental movements leads to ignoring the interaction between body segments. Ae and Shibukawa (1980) and Lees and Barton (1996) divided segment momentum into two components: transfer

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momentum and relative momentum. They used the relative momentum to represent the contribution of arms to vertical momentum. These studies are interesting in that the arm swing effect was discussed as the relative momentum of the arm to the shoulder. However, it was not taken into consideration whether the movement of the lower extremities was changed by the arm swing.

Feltner et al. (1999) studied the change of the resultant joint torque at the hip, knee and ankle during countermovement vertical jumps with and without an arm swing using inverse dynamics. They showed that during the final two-thirds of the propulsive phase, the arm swing augmented hip extensor torques by slowing the rate of trunk extension and placing the hip extensor muscles in slower concentric conditions that favored the generation of larger forces and resultant joint torques. The significance of their study is the discovery that the arm swing had some influence on the lower extremity movement. But they did not discuss the work output by each joint, and therefore, the problem of quantifying the arm-swing effect remains to be solved.

Recently, Lees et al. (2004) examined the theories and mechanisms that underpin enhanced performance in the vertical jump when using an arm swing. In their experiment, subjects jumped higher (0.086 m) in the arm swing compared to the no-arm swing condition and they showed that this increase was due to increased height (28%) and velocity (72%) of the CM at take-off. The increased height at take-off was due to the elevation of the arm segments. The increased velocity at take-off stemmed from a complex series of events, which allowed the arms to build up energy early in the jump and transfer it to the rest of the body during the later stages of the jump. This energy came from the shoulder and elbow joints as well as from extra work done at the hip.

According to the work–energy theorem, the sum of work done by each joint is equal to: potential energy of CM + kinetic energy of CM + relative kinetic energy of each segment about CM. The essence of vertical jump movement is to transform joint work into kinetic energy of vertical movement and/or potential energy. It is important to examine the influence of the arm swing on

both individual joint work and on total body work. Despite that, the effect of arm swings has not been sufficiently discussed by researchers.

The purpose of this study, therefore, was to quantify how the arm swing affects lower extremity torque, power and work during the vertical jump and to gain an insight into the mechanism that enables the arm swing to increase the jump height.

2. Methods

2.1. Experimental procedure

Five healthy male subjects (age: 26.3 ± 3.2 year, height: 172.0 ± 8.9 cm, body mass: 69.6 ± 6.3 kg; mean $\pm$ SD) participated in the experiment. The subjects read a description of the purpose, risks and basic procedures of the experiment and then gave their written informed consent to participate. This study was approved by the Ethics Committee of the University of Tokyo. The subjects were marked with reflective markers at the anatomical landmarks necessary for the linked segment analysis: the joint centers of the wrist, elbow, shoulder, hip, knee and ankle, and the fifth metatarsal–phalangeal joint on the right side of the body. After adequate warming up and practice, they maximally jumped vertically from a force platform (9281B, Kistler, Switzerland) two different ways: squat jump with no arm swing (SJ) and squat jump with an arm swing (SJA) as shown in Fig. 1. Before the trials, we recorded each subject performing a squat jump in their preferred way using a high-speed camera (MEMRECAMc²s, NAC, Japan). The recorded motions were replayed on a TV monitor and the joints of the lower limb (ankle, knee and hip) at the starting position were marked on the monitor with a pen. These marks were used as reference position for each subject to initiate both SJ and SJA. This process was needed to make the intraindividual lower body posture of SJ and SJA the same within each subject. In SJ, the hands remained on the subject's iliac crests to restrict arm movement (Fig. 1). For SJA, the subject's arms were extended behind the back, held

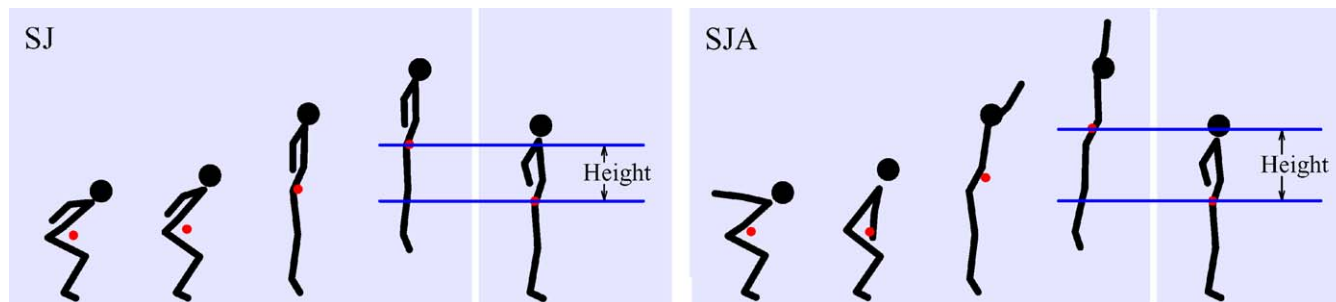


Fig. 1. Stick pictures for two types of jump. SJ; squat jump with no arm swing, SJA; squat jump with arm swing. The jump height was defined as the maximum displacement of the CM of the body (the dots in the figures) from the standing position.

motionless, then swung forward at each subject's preferred timing during the jumping motion. All the subjects performed four trials per each jump condition. The order of jumps was randomized and two-minutes of rest was allowed between each trial.

2.2. Measurement

The vertical and horizontal components of the ground reaction forces ($\mathbf{F}_{GR}$) and cinematographic data of each jump were simultaneously collected at 200 Hz using a square wave pulse electronic start switch. We measured the kinematics of the right side of the subjects using a high-speed camera (Fig. 2). The camera was located 7.2 m from the subjects with its optical axis oriented perpendicular to the sagittal plane of the subjects. The camera provided a field of view ($\sim 1.5 \text{ m} \times 2.5 \text{ m}$) large enough to record the motions of the participants. Prior to the subjects performing the vertical jumps, a calibration board ($1.0 \text{ m} \times 2.0 \text{ m}$) placed on the force platform perpendicular to the camera's optical axis was videotaped. Thirty reflective markers (diameter: 1.0 cm) were put on known locations of the board. The control signals of the camera and the output of the force platform were simultaneously recorded by a personal

computer so that the force and kinematic data could be synchronized.

2.3. Analyses

2.3.1. Center of pressure and the displacement of the CM

The vertical and horizontal components of $\mathbf{F}_{GR}$ and the center of the pressure (COP) were obtained from the force platform data (Nigg and Herzog, 1995). The displacement of the CM of the body at time t was calculated as follows:

$$\mathbf{V}_{CM}(t) = \frac{1}{M} \int_{t_0}^t (\mathbf{F}_{GR} + M\mathbf{g}) dt,$$

$$\Delta \mathbf{R}_{CM}(t) = \int_{t_0}^t \mathbf{V}_{CM}(t) dt,$$

where M is the body mass, t_0 the time slightly before the beginning of the motion, $\mathbf{g}$ the acceleration of gravity, $\mathbf{V}_{CM}(t)$ the velocity of the CM and $\Delta \mathbf{R}_{CM}(t)$ the displacement of the CM. The jump height was defined as the maximum displacement of the CM from the standing position (Fig. 1). We calculated the jump height of the four trials per each jump condition and the highest jump performance for each condition was used for the subsequent analyses.

2.3.2. Calibration and calculation of the kinematics

The centers of the calibration points were first digitized. The data obtained from digitizing the images of the calibration points together with their known locations were used to calculate the direct linear transformation (DLT) parameters for the camera (Abdel-Aziz and Karara, 1971). The digitized coordinate data for the jump performances along with the camera parameters were used to reconstruct the two-dimensional locations of the joint centers using the DLT method. All digitizing and the DLT method were carried out with the Frame-DIAS Digitizing and Calibration System (DKH Co., Japan). The obtained time course coordinate data were smoothed with a low-pass filter (fourth-order zero-lag Butterworth filter, cut-off frequency 5 Hz; Winter, 1990). Using a link segment model, we calculated the kinematic variables (angle, angular velocity, angular acceleration) of each joint (ankle, knee, hip, shoulder, elbow) from the beginning of the motion to the takeoff. Fig. 2 displays how each joint angle was defined.

2.3.3. Torque

The equation of motion of the segment CM and the equation of rotational motion about each segment CM were used to calculate each joint reaction force and torque (inverse dynamics method: Winter, 1990;

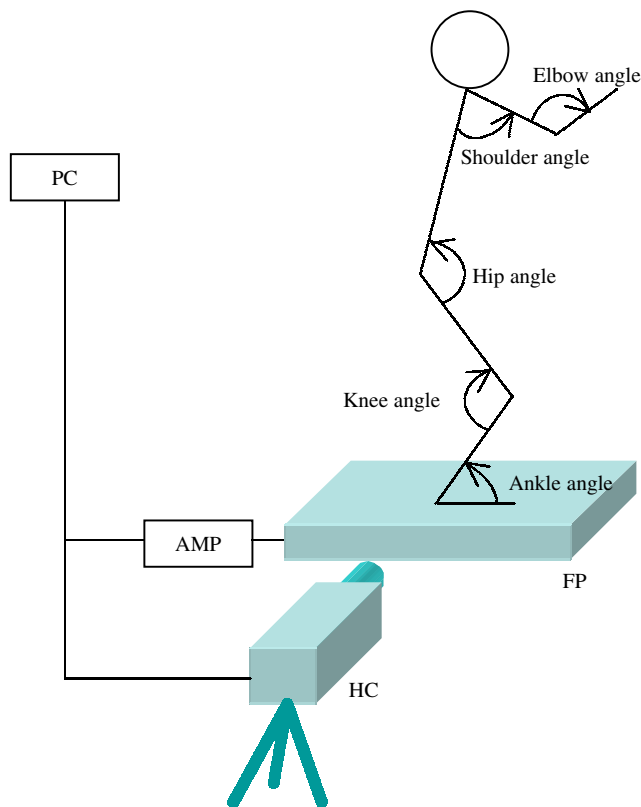


Fig. 2. Joint angle definitions and the experimental setting (PC; personal computer, AMP; amplify, HC; high speed camera, FP; force plate). The arrows show the positive direction of each joint angle.

Table 1
Variables in the equations of the inverse dynamics method in 2.3.3

m	:	mass of the segment
I_G	:	moment of inertia created about the segment CM
$\dot{\omega}$	:	angular acceleration of the segment
$\mathbf{a}_{CM}$	:	acceleration of the segment CM
$\mathbf{g}$	:	acceleration of gravity
$\mathbf{R}\mathbf{J}\mathbf{F}_{PROXIMAL}$	:	proximal resultant joint force exerted on the segment
$\mathbf{F}_{DISTAL}$	:	distal force applied to the segment
$\mathbf{R}\mathbf{J}\mathbf{T}_{PROXIMAL}$	:	proximal resultant joint torque
$I_G \dot{\omega}$	:	rate of change of the angular momentum with respect to time about the segment CM (= the resultant torque about the segment CM)
$\mathbf{R}\mathbf{J}\mathbf{T}_{DISTAL}$	:	distal resultant joint torque applied to the segment
$T_{F\ PROXIMAL}$	:	torque created about the segment CM by the proximal resultant joint forces applied to the segment
$T_{F\ DISTAL}$	:	torque created about the segment CM by the distal resultant joint forces applied to the segment

Feltner et al., 1999). The equations are as follows:

$$\mathbf{R}\mathbf{J}\mathbf{F}_{PROXIMAL} = m(\mathbf{a}_{CM} - \mathbf{g}) - \mathbf{F}_{DISTAL},$$

$$\mathbf{R}\mathbf{J}\mathbf{T}_{PROXIMAL} = I_G \dot{\omega} - \mathbf{R}\mathbf{J}\mathbf{T}_{DISTAL} - T_{F\ PROXIMAL} - T_{F\ DISTAL}.$$

The variables in these equations are explained in Table 1. The equations for the feet, legs, thighs, forearms and arms were used to calculate the joint reaction force and torque at the ankle, knee, hip, elbow and shoulder joints, respectively. The mass of each segment and the moment of inertia values about the transverse axis of each segment reported by Ae (1996) were used and personalized for each participant.

2.3.4. Power and work

Each joint power at time t was given by

$$P_i(t) = T_i(t) \times \omega_i(t),$$

where T_i is the resultant joint torque and ω_i is the joint angular velocity. Work done by each joint throughout the motion was calculated by integrating the power with respect to time:

$$W_i = \int_{t_0}^0 P_i(t) dt.$$

In these equations, the subscript i is the index of each joint. The instant of takeoff was chosen as $t = 0$, and t_0 is the time slightly before the beginning of the motion. We examined the ground reaction forces of all the performances and determined $t_0 = -0.6$ s.

The lower extremity work (W_L), the upper extremity work (W_U) and the total body work (W_T) were calculated as follows:

$$W_L = W_A + W_K + W_H,$$

$$W_U = W_S + W_E,$$

$$W_T = W_L + W_U,$$

where each subscript is A: ankle joint, K: knee joint, H: hip joint, S: shoulder joint and E: elbow joint.

Mean curves of variables shown in the Results section were calculated for the group of subjects after synchronization of individual curves from the instant the toes lost contact with the force platform ($t = 0$).

2.4. Statistics

A one-sided paired t -test was used to compare between SJ and SJA for jump height, displacement and velocity of the CM at takeoff, total body work, each joint work, lower extremity work and upper extremity work.

3. Results

3.1. Comparison between SJ and SJA

Table 2 shows the kinematic and kinetic values of SJ and SJA. There were significant differences between SJ and SJA in the displacement and velocity of the CM (D_{CM} and V_{CM}) at takeoff ($p < 0.01$). D_{CM} at takeoff could not directly be compared between our study and the preceeding studies of Harman et al. (1990), because our study defined D_{CM} as the distance from a standing position, while Harman et al. defined the CM displacement as the distance from a squat position. Additionally, the initial squat position was not controlled in their experiments. Therefore, displacement in their studies would have varied depending on the selected starting squat height in each trial. By ensuring that each subject's trials started from the same squat position, we have tried to eliminate that variation. The values of V_{CM} at takeoff in SJ and SJA were 2.30 ± 0.28 and 2.53 ± 0.31 in Harman et al. (1990). The values of V_{CM} at takeoff in CJ (countermovement jump with no arm swings) and CJA (countermovement jump with arm swings) were 2.58 ± 0.14 and 2.81 ± 0.14 in Lees et al. (2004), and

Table 2

Comparison between SJ and SJA for the jump height, D_{CM} at takeoff, V_{CM} at takeoff, $F_{GR,Z}$ peak, $F_{GR,Z}$ impulse, W_A , W_K , W_H , W_S , W_L , W_U , W_T

	SJ	SJA	<i>t</i> -test
Jump height (10^{-1} m)	4.45 ± 0.65	5.46 ± 0.49	$p < 0.01$
D_{CM} at takeoff (10^{-1} m)	1.54 ± 0.15	1.87 ± 0.24	$p < 0.01$
V_{CM} at takeoff ($m s^{-1}$)	2.46 ± 0.27	2.73 ± 0.20	$p < 0.01$
$F_{GR,Z}$ peak (10^3 N)	1.59 ± 0.16	1.73 ± 0.17	$p < 0.01$
$F_{GR,Z}$ impulse (102 N s)	1.72 ± 0.20	1.90 ± 0.14	$p < 0.01$
W_A (10^2 J)	1.39 ± 0.27	1.60 ± 0.30	$p < 0.01$
W_K (10^2 J)	1.24 ± 0.38	0.98 ± 0.21	$p < 0.05$
W_H (10^2 J)	1.98 ± 0.63	2.34 ± 0.40	$p < 0.05$
W_S (10^2 J)	—	0.15 ± 0.09	—
W_E (10^2 J)	—	0.02 ± 0.02	—
W_L (10^2 J)	4.62 ± 0.85	4.93 ± 0.72	$p < 0.05$
W_U (10^2 J)	—	0.16 ± 0.01	—
W_T (10^2 J)	4.62 ± 0.85	5.09 ± 0.76	$p < 0.01$

2.44 ± 0.23 and 2.75 ± 0.28 in Feltner et al. (1999), respectively. Although the Feltner study used counter-movement, V_{CM} at take-off in our results were quite similar to those of Feltner et al.'s (1999). As there were some deviations in values, the four experiments including ours commonly showed that V_{CM} at take-off was larger in SJA than in SJ (CJA than in CJ).

Fig. 3 shows ensemble averages ($n = 5$) of the z component of F_{GR} ($F_{GR,Z}$) for SJ and SJA. In both conditions, the force began to rise at slightly after $t = -0.6$ s. Therefore, we defined the push-off phase as the period between $t = -0.6$ s and takeoff. Not only the peak value of $F_{GR,Z}$ but also the impulse of $F_{GR,Z}$ was larger in SJA than in SJ. This resulted in the larger velocity of CM at takeoff in SJA.

There was a significant difference in the jump heights between SJ and SJA ($p < 0.01$, Table 2). This result showed that an arm swing in a squat jump enhances the jump height, which is consistent with prior studies (Payne et al., 1968; Harman et al., 1990).

There was a significant difference between SJ and SJA in the total work of all the joints ($p < 0.01$, Table 2). This increase of the total body work by an arm swing (ΔW_T) was the direct cause of the rise of the jump height.

The total work by the lower extremity joints (ankle, knee and hip) and the total work by the upper extremity joints (shoulder and elbow) were compared respectively between SJ and SJA in Table 2. The enhancement (SJA–SJ) of the work by the lower extremity joints (ΔW_L) and the upper extremity joints (ΔW_U) was 31.4 and 16.3 J, respectively. The ratios of the enhancement of the lower and upper extremity work to the increase of the total body work ($\Delta W_L/\Delta W_T \times 100$, $\Delta W_U/\Delta W_T \times 100$) were 65.9% and 34.1%, respectively. These results suggest that the increment of the jump height resulted mainly from the change of the lower extremity work. If we divide the increment of the jump height (10.1 cm) in proportion to the ratio of work, we can estimate that

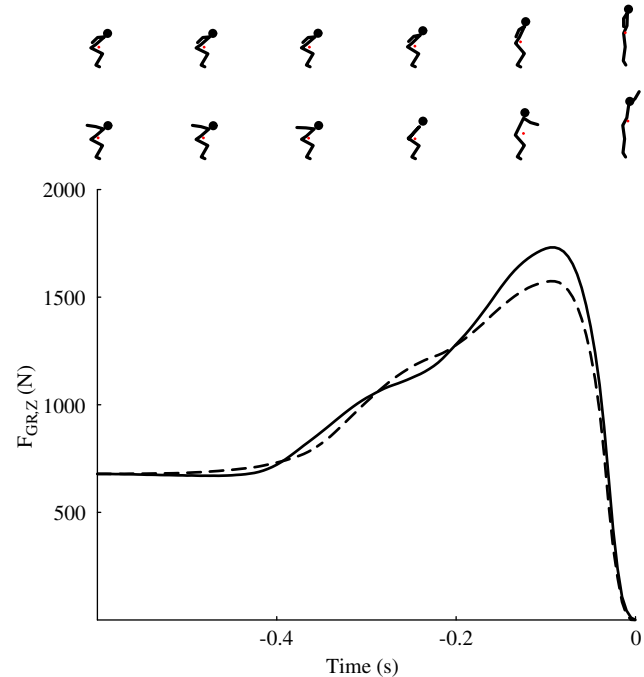


Fig. 3. Ensemble averages ($n = 5$) of $F_{GR,Z}$ for SJ (dashed line) and SJA (solid line). Zero on the time axis indicates the takeoff.

3.4 cm increment was due to the upper extremity work and 6.7 cm to the lower extremity work. The increment of the CM displacement by raising the arms relative to the trunk can be calculated as 3.3 cm (relative displacement of the CM of the arm $\times$ mass of the arm/total body mass). These results suggested that the increment of the work in upper extremities was mostly used to raise the arms.

Ensemble averages ($n = 5$) of each joint (ankle, knee, hip, shoulder and elbow) work for SJ and SJA were shown in Table 2. Work by the ankle and by the hip was significantly larger in SJA than in SJ ($p < 0.01$). The ratios of the enhancement of the ankle work and hip work were 14.4% and 18.6% ($\Delta W_j/SJW_j \times 100$, suffix j indicates each joint), respectively. The work by the knee was significantly smaller in SJA than in SJ (-20.5% , $p < 0.05$). On the other hand, the ratios of the enhancement calculated from the data in Lees et al.'s study (2004) were 14.1%, 0.014% and 0.0% for hip, ankle and knee work, respectively. The difference between their study and ours maybe resulted mainly from the difference in the models of movement: we used squat jumps while they used countermovement jumps.

3.2. Time history of each mechanical variable

The angle and angular velocity for lower extremities joints are shown in Fig. 4, and the torque, power and work for each joint are shown in Fig. 5.

The angle and angular velocity curves of SJ were almost identical to those of SJA for the ankle and knee

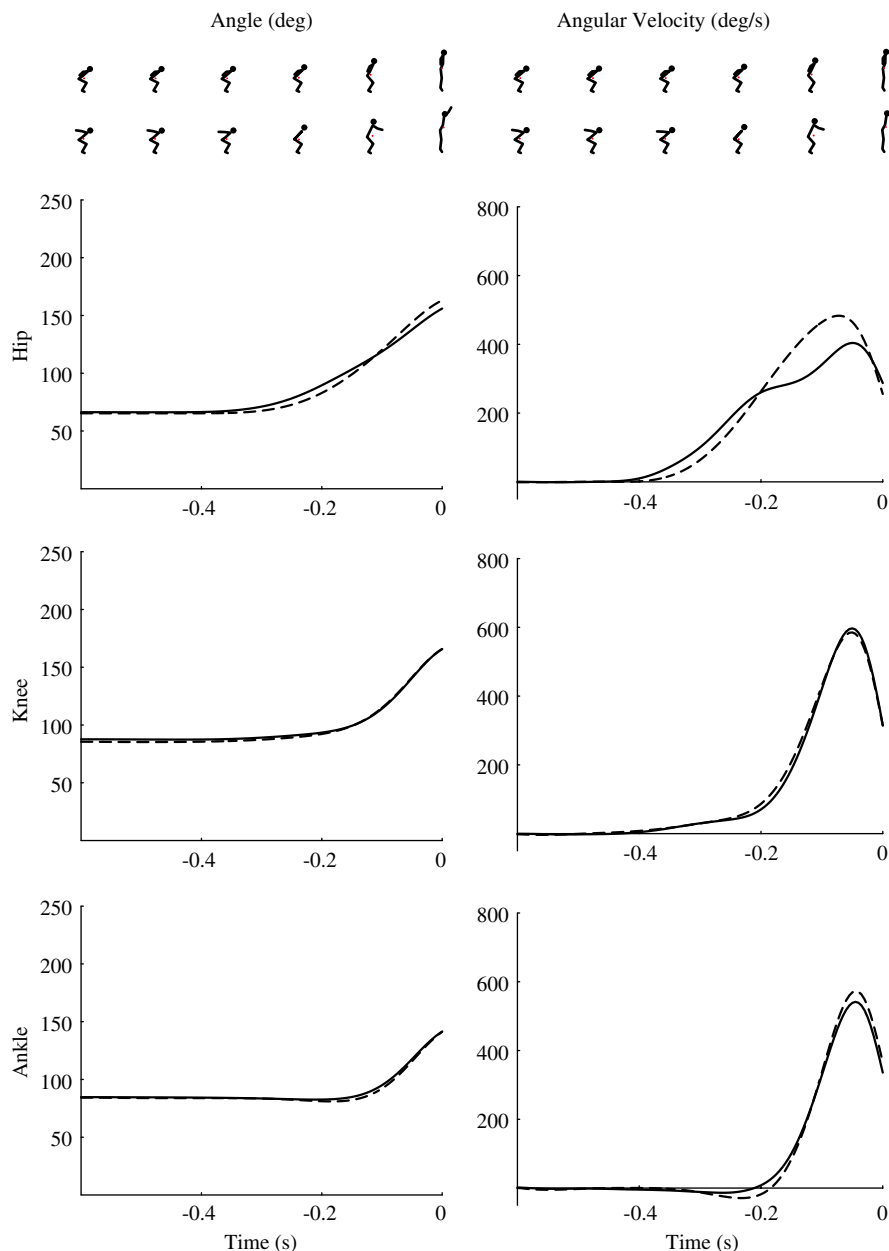


Fig. 4. Ensemble averages ($n = 5$) of joint angle (left) and angular velocity (right) for SJ (dashed line) and SJA (solid line). The positive direction was set for hip extension, knee extension and planter flexion.

joints. For the hip joint, the angular velocity reduced in the latter half of the push-off phase in SJA (from $t = -0.2$ s to $t = -0.05$ s). The peak value of the hip joint torque in SJ was nearly equal to that in SJA. However, before and after the peak, SJA generated larger torque. Although the knee joint torque did not differ in the peak value between SJA and SJ, it was smaller in SJA than in SJ from $t = -0.4$ s to $t = -0.15$ s. For the ankle joint, both the peak value of the torque and the angular impulse were larger in SJA than in SJ.

The power by the hip joint was greater in SJA than in SJ except around its peak. As a result, the work by the hip joint was larger in SJA than in SJ. The power by the

ankle joint was greater in SJA than in SJ throughout the push-off phase, which resulted in the increment of ankle joint work in SJA. On the contrary, power by the knee joint was greater in SJ than in SJA throughout the push-off phase, which resulted in the decrease of knee joint work in SJA.

4. Discussion

We found in the previous section that the total work done by all the joints increased with an arm swing. 34% of this increase came from the upper extremity joints

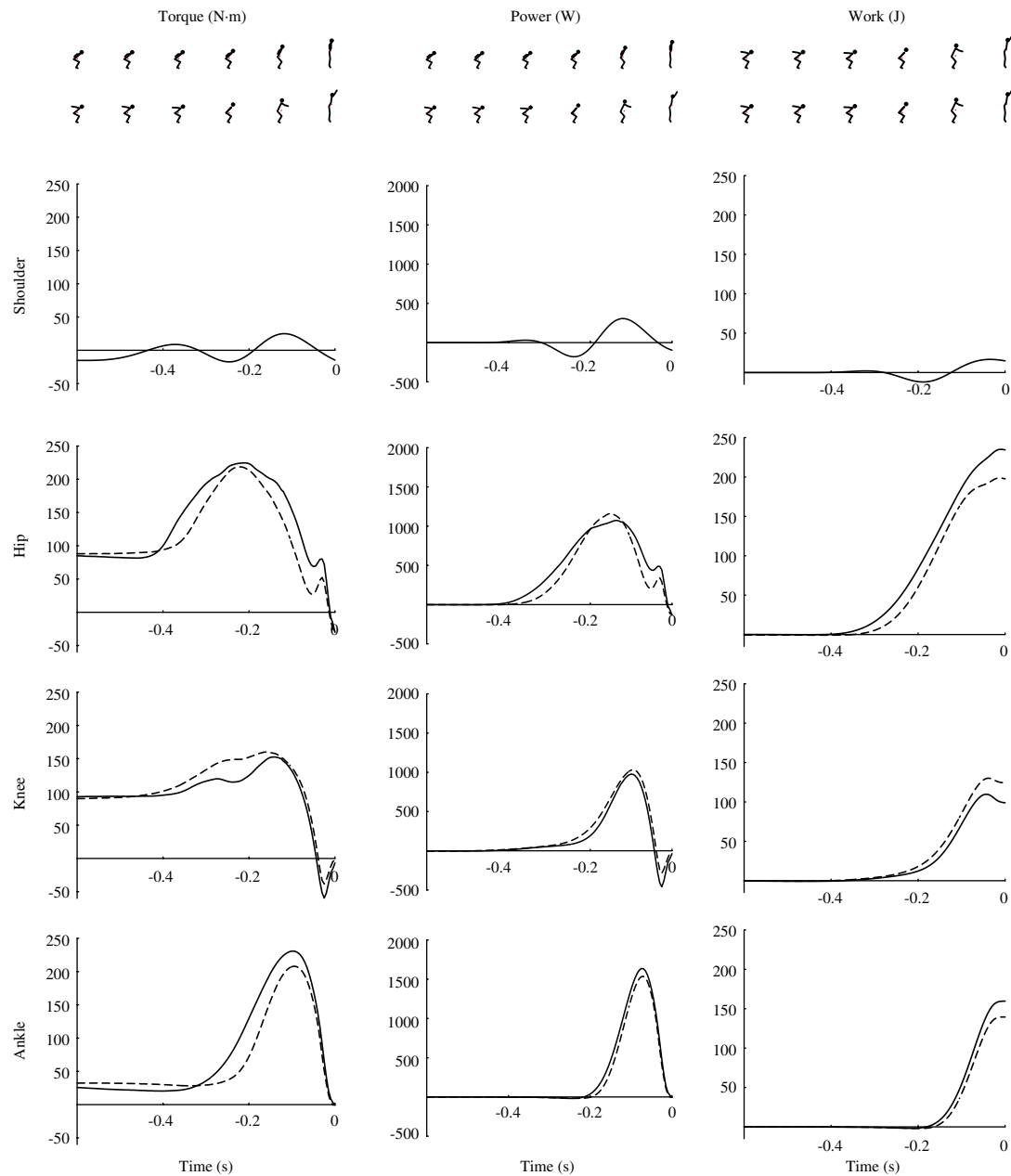


Fig. 5. Ensemble averages ($n = 5$) of torque (left), power (middle), and work (right) for SJ (dashed line) and SJA (solid line). The positive direction was set for shoulder flexion, hip extension, knee extension and planter flexion.

and 66% from the lower extremity joints. In the following sections, we have focused on factors influencing the increase of the lower extremity work in the squat jumping.

4.1. Factors influencing the increase of hip joint work

Work by the hip was significantly larger in SJA than in SJ. The arm swing in vertical jumping acts as an additional load to tilt the trunk forward. This load is divided into two components: the shoulder joint torque applied to the trunk (T_S) and the torque created about

the CM of the trunk by the shoulder joint force applied to the trunk (T_F).

As shown in Fig. 5, the value and timing of the peak hip-joint torque in SJ were almost equal to those in SJA. However, before and after the peak, the torque was larger in SJA, which resulted in the increase of the hip joint work. The hip-torque increment in the first half of the push-off phase in SJA can be explained as follows: Because the hip extension took a longer time due to the additional load ($T_S + T_F$) by the arm swing, the movement of the hip joint started earlier, i.e. the hip joint angle, angular velocity and torque began to rise earlier in SJA than in SJ.

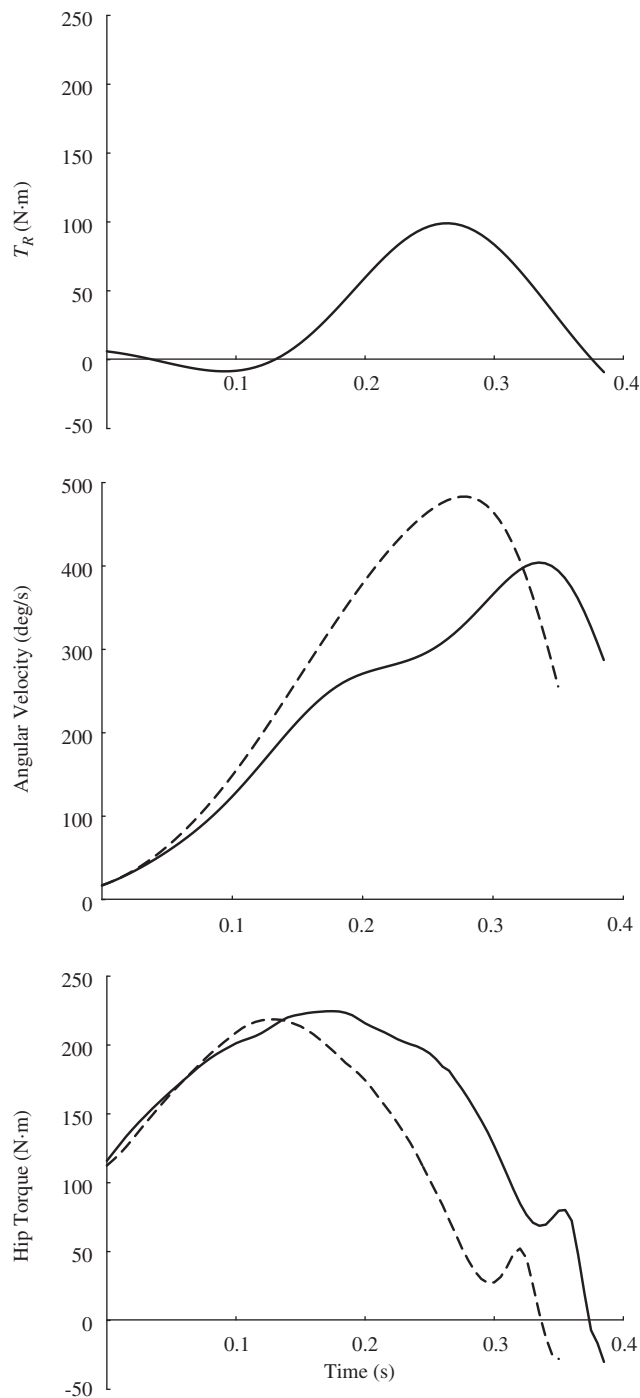


Fig. 6. Top: contribution of the arm swing to the hip joint torque; T_R (resultant load). Middle: hip angular velocity; SJ (dashed line), SJA (solid line). Bottom: hip joint torque; SJ (dashed line), SJA (solid line). The origin of the time coordinate was set at the beginning of the increase of the hip angular velocity in both SJ and SJA.

Fig. 6 shows T_R (resultant load: $T_S + T_F$) in SJA, and the angular velocity and torque of the hip joint in SJ and in SJA. In this figure, the origin of the time coordinate was set at the beginning of the increase of the hip angular velocity in SJ and SJA, respectively. The hip

joint torque was augmented in the period when T_R was large. Feltner et al. (1999) explained that this phenomenon was due to the slower extension of the hip joint caused by T_R (Hill's force–velocity relationship: Hill, 1938). Feltner's suggestion may be one of the factors for the hip torque augmentation. T_R became larger in the latter half of the push-off phase, and in this period, the angular velocity was smaller and the hip joint torque was larger in SJA than in SJ. However, we have to keep in mind that the force–velocity relationship holds true on condition that the muscle activation level is kept the same. None of the muscles are activated maximally all the time in jumping even if the subject performs a maximal jump. As activation level may vary according to the jump conditions, it is reasonable to suggest that in SJA the muscle activation level increased in the latter half of the push-off phase compared with SJ, which resulted in the increase in the muscle torque. The examination of EMG data in future studies is therefore warranted.

4.2. Factors influencing the increase of ankle joint work

We have shown that work by the ankle was significantly larger in SJA than in SJ ($p < 0.01$). The ankle angle and angular velocity curves of SJ were almost identical to those of SJA (Fig. 4). On the other hand, the torque at the ankle joint was larger in SJA during the push-off phase, which contributed to the increase of the ankle work (Fig. 5). The force–velocity relationship alone cannot account for this increase in torque. We suggest that the muscle activation levels themselves are different between SJA and SJ. Since muscle output is regulated in order to jump vertically, not all muscles are activated maximally even if the subject performs a maximal jump. Therefore, muscles still have some capacity left for higher activation. However, the local (imbalanced) increase in joint torques cannot be allowed because the task is restricted. The swinging of the arms, however, changes the posture of the body and, in doing so, likely changes the coordination of the muscle activation. Plantar flexors may well be more activated against a downward load to maintain the posture to jump vertically. This is probably one reason for the increase in ankle torque. Another factor for the enhanced work around the ankle might be transported energy from the knee extensors to the ankle joint via the bi-articular gastrocnemius.

4.3. Factors influencing the increase of knee joint work

The knee angular velocity in SJA was almost the same as that in SJ throughout the push-off phase (Fig. 4). On the other hand, in the period between -0.4 and -0.1 s, the knee torque in SJA was smaller than that in SJ (Fig. 5). The decrease of the knee torque was the

direct cause of the decrease of the knee work. The decrease of the knee torque may be due to the effect of the biarticular muscles. As discussed in the section on the hip joint work, larger hip torque was exerted in SJA. This suggests that the hip extensors were more activated when the arm swing was performed. An increased activation of the biceps femoris, which acts not only as a hip joint extensor but also as a knee flexor, would reduce knee extension torque. A similar reasoning can be applied to the gastrocnemius muscle. As discussed in the previous section on the ankle joint work, the plantar flexion torque was augmented by the arm swing. Knee extension torque could have been reduced by the biarticular gastrocnemius muscle, which acts both as a plantar extensor and as a knee flexor. Although active knee torque is lowered, the knee extension can be facilitated by the increased hip extension torque and plantar flexion torque, preventing a decrease in knee angular velocity.

5. Conclusion and perspective

Our study focused on mechanical aspects of vertical jumps with and without an arm swing. The increment of the jump height with an arm swing was the result of the increase of the total body work, and our study revealed that the greater part of this increase comes from the lower extremity joints. Of the total increase in joint work, 34.1% originated from the upper extremity joints and 65.9% originated from the lower extremity joints.

One of the factors for the increase of the hip joint work may be the force–velocity relationship as was indicated by Feltner et al. (1999). However, activation level may vary according to the jump conditions, which can also contribute to the increase of the work. Work at the ankle was also augmented although no clear decrease in ankle plantar flexion velocity was observed. This was also probably due to the changes in the muscle activation level. Work at the knee was reduced, which might be connected to the transport of energy of bi-articular muscles to the hip and the ankle. Further study using EMG data would be useful in determining individual muscle output in different jumping conditions.

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